CATASTROPHE • 2011

Cyclone Yasi

The night the Wet Tropics went dark — and a World Heritage rainforest fell silent.

Mission Beach, QLD • 2011

FAST FACTS

1	DIRECT FATALITY	**$3.6bn**	TOTAL DAMAGE (2011)
Cat 5	SEVERE CYCLONE	**929 hPa**	CENTRAL PRESSURE
300+	PEAK GUSTS KM/H	**175,000**	PEOPLE EVACUATED

TIMELINE

29 Jan 2011
Tropical low develops east of Vanuatu.

1 Feb, Afternoon
Rapidly intensifies to Category 5; 285 km/h sustained.

2 Feb, ~Midnight
Eye crosses coast near Mission Beach — most intense QLD landfall since 1918.

2 Feb, Morning
Storm weakens over Cape York Peninsula.

On the night of 1 February 2011, residents across Far North Queensland had already packed what they could carry and driven inland. The Bureau of Meteorology had been unambiguous for two days: Cyclone Yasi was going to be the most powerful storm to strike the Queensland coast in living memory. The warnings worked. But no warning could prepare the land itself.

At approximately midnight AEST, Yasi’s eye crossed the coast between Mission Beach and Cardwell as a Category 5 severe tropical cyclone. Sustained winds of 285 km/h, gusting beyond 300 km/h, tore through the coastal strip. A storm surge of up to five metres inundated the beachfront. The Wet Tropics World Heritage Area took the full force of the eye wall.

> *“We thought we understood these forests. They’re hundreds of millions of years old. But when we walked in after Yasi, it was like walking on another planet. The canopy was simply gone.”*
> **— Dr. Erin Vandermark, Wet Tropics Authority, February 2011**

A Storm of Record Force

Yasi’s gale-force wind radius extended over 650 km — simultaneously affecting Cairns to the north and Townsville to the south, covering an area greater than the United Kingdom. An estimated 175,000 people evacuated. The death toll — one direct fatality — reflected what can be achieved when warnings are clear and heeded.

Wet Tropics rainforest near Mission Beach following Cyclone Yasi. Ancient trees snapped at the trunk. The canopy, centuries in the making, was stripped in hours.

The Agricultural Reckoning

Just five years after Larry obliterated the banana crop, Yasi struck the same region. An estimated 75% of Australia's banana crop was again destroyed. Sugar cane fields across Tully and Innisfail were flattened. Total economic damage reached A$3.6 billion.

STORM SURGE

Storm surge inundation at Mission Beach — Yasi's surge reached up to five metres along the coastal strip, devastating beachfront communities.

■ LEARNING FOCUS: Ecosystem Vulnerability & Recovery

Natural disasters reveal how ecosystems respond to sudden, extreme stress.

- ✓ **World Heritage Areas** are places of extraordinary natural value protected under international law. The Wet Tropics harbours 3,000+ plant species and 107 mammal species.
- ✓ **Ecological resilience** is the ability to absorb disturbance and reorganise. Ancient rainforests have evolved to recover from cyclones over millions of years.
- ✓ **Climate change** is increasing the frequency and intensity of Category 4 and 5 cyclones. If ecosystems face more frequent extreme events, they may not recover between each one.
- ✓ **Coral bleaching** occurs when stress causes coral to expel its algae. Cyclone runoff, combined with warming oceans, puts reef systems at serious long-term risk.